MIDDLESEX SCHOOL | MATHEMATICS

Math 21 Course at a Glance

Algebra and Its Functions

Topical units are not numbered because instructional order may vary. Objectives are grouped by curriculum priority; supporting skills are intentionally omitted from this summary.

Review Content

REVIEW	
M21-REV-001	I can determine and interpret the slope of a line.
M21-REV-002	I can graph a line from an equation and write an equation from given information.
M21-REV-003	I can apply a linear function to a context.
M21-REV-004	I can identify the solution of a linear system as an intersection point.
M21-REV-005	I can solve a simple two-variable linear system using an appropriate method.
M21-REV-006	I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED	
M21-FUN-001	I can evaluate a function from a formula, graph, or table.
M21-FUN-002	I can determine whether a relation is a function.
M21-FUN-003	I can determine domain and range from a graph or table.
M21-FUN-004	I can solve for an input that produces a specified output.
M21-FUN-005	I can interpret inputs, outputs, domain, and range in context.
M21-FUN-006	I can calculate and interpret average rate of change.
M21-FUN-007	I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED	
M21-POL-001	I can add and subtract polynomials.
M21-POL-002	I can multiply polynomials.
M21-POL-003	I can factor out a greatest common factor.
M21-POL-004	I can factor a quadratic trinomial.
M21-POL-005	I can factor a difference of squares and other required special cases.
M21-POL-006	I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED	
M21-RAT-001	I can state excluded values for a rational expression.
M21-RAT-002	I can simplify a rational expression while preserving its restrictions.
M21-RAT-003	I can multiply and divide rational expressions.
M21-RAT-004	I can add and subtract rational expressions using a common denominator.
M21-RAT-005	I can solve a rational equation and reject extraneous solutions.

Exponents and Radicals

REQUIRED	
M21-EXP-001	I can simplify expressions containing negative exponents.
M21-EXP-002	I can rewrite expressions between radical and rational-exponent forms.
M21-EXP-003	I can simplify radical expressions.
M21-EXP-004	I can apply exponent rules to expressions with integer and rational exponents.
M21-EXP-005	I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

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Quadratic Functions and Equations

REQUIRED

M21-QUAD-001	I can identify key features of a quadratic function from its graph or equation.
M21-QUAD-002	I can connect standard, factored, and vertex forms to features of a parabola.
M21-QUAD-003	I can solve a quadratic equation by factoring.
M21-QUAD-004	I can solve a quadratic equation using the square-root property.
M21-QUAD-005	I can solve a quadratic equation by completing the square.
M21-QUAD-006	I can solve a quadratic equation using the quadratic formula.
M21-QUAD-007	I can choose an efficient method for solving a quadratic equation.
M21-QUAD-008	I can graph a quadratic function using its vertex, intercepts, and symmetry.
M21-QUAD-009	I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION

M21-EXT-001	I can perform basic arithmetic with complex numbers.
M21-EXT-002	I can express the nonreal solutions of a quadratic equation using complex numbers.